PSTAT126 Final Project

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Perm Number: A1C7R60 | *Section:* Wednesday 2–2:50 PM

Project Instructions: Final_Project.pdf

PART 1 — Data Description and Descriptive Statistics

1) Select a Random Sample

```
setwd("C:/Users/jaket/Downloads/PSTAT126")

# Reading in .csv file
diamond_data <- read.csv("Diamonds Prices2022.csv")

# Set seed for reproducibility
set.seed(1)

# Select a Random Sample (Size n = 1000)
diamond_sample <- diamond_data[sample(nrow(diamond_data), 1000), ]
```

2) Describe All Variables

```
# Dataset Summary
summary(diamond_sample)
```

```
##        X             carat            cut                color
##  Min.   :   40   Min.   :0.2300   Length:1000        Length:1000
##  1st Qu.:11924   1st Qu.:0.4000   Class :character   Class :character
##  Median :26047   Median :0.7150   Mode  :character   Mode  :character
##  Mean   :26263   Mean   :0.8353
##  3rd Qu.:39308   3rd Qu.:1.0725
##  Max.   :53917   Max.   :5.0100
##    clarity              depth           table           price
##  Length:1000        Min.   :55.60   Min.   :52.00   Min.   :  362
##  Class :character   1st Qu.:61.10   1st Qu.:56.00   1st Qu.:  957
##  Mode  :character   Median :61.90   Median :57.00   Median : 2678
##                     Mean   :61.82   Mean   :57.45   Mean   : 4186
##                     3rd Qu.:62.60   3rd Qu.:59.00   3rd Qu.: 5420
##                     Max.   :70.20   Max.   :70.00   Max.   :18757
##        x                y               z
##  Min.   : 0.000   Min.   : 0.000   Min.   :0.000
##  1st Qu.: 4.737   1st Qu.: 4.730   1st Qu.:2.920
```

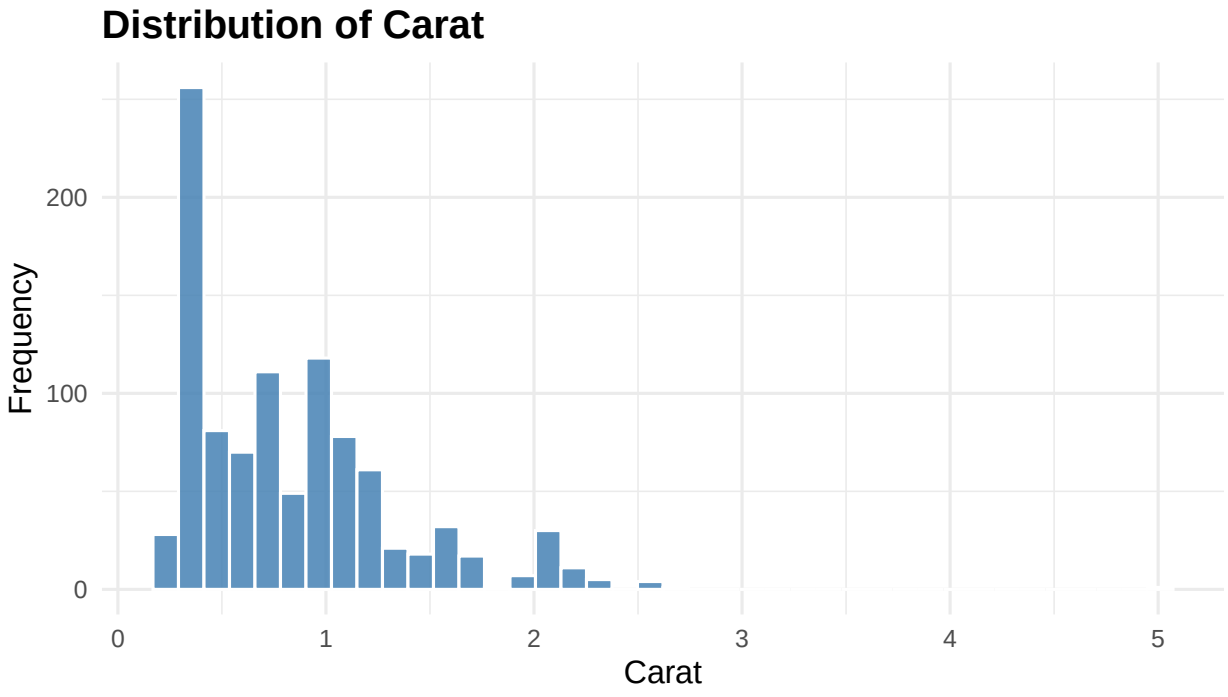
```
##  Median : 5.750   Median : 5.790   Median :3.550
##  Mean   : 5.797   Mean   : 5.801   Mean   :3.581
##  3rd Qu.: 6.572   3rd Qu.: 6.582   3rd Qu.:4.060
##  Max.   :10.740   Max.   :10.540   Max.   :6.980
```

```
# Dataset Structure
str(diamond_sample)
```

```
## 'data.frame':    1000 obs. of  11 variables:
##  $ X      : int  24388 43307 4050 11571 25173 32618 50951 13903 8229 25305 ...
##  $ carat  : num  0.41 0.5 1.03 1.1 1.51 0.3 0.87 1.05 1 2.01 ...
##  $ cut    : chr  "Very Good" "Very Good" "Fair" "Ideal" ...
##  $ color  : chr  "D" "F" "I" "I" ...
##  $ clarity: chr  "SI2" "VS2" "SI2" "SI1" ...
##  $ depth  : num  62.3 62.8 65.2 62.1 63.3 62.1 61.4 63.3 64 63.8 ...
##  $ table  : num  61 57 56 57 61 55 57 57 57 57 ...
##  $ price  : int  638 1402 3530 5037 13757 457 2321 5657 4372 13976 ...
##  $ x      : num  4.72 5.05 6.42 6.6 7.24 4.3 6.17 6.45 6.29 7.95 ...
##  $ y      : num  4.75 5.08 6.35 6.64 7.17 4.33 6.14 6.4 6.33 7.91 ...
##  $ z      : num  2.95 3.18 4.16 4.11 4.56 2.68 3.78 4.07 4.04 5.06 ...
```

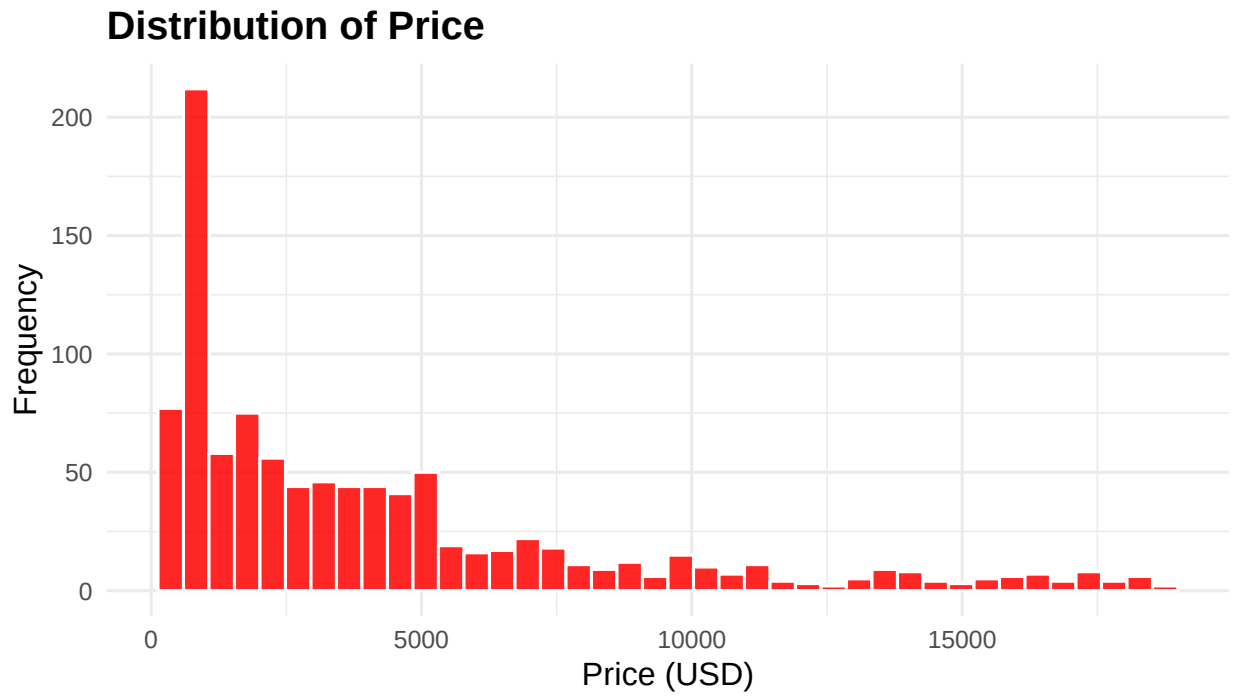
Continuous Variables

```
# Carat - Diamond Weight (1 Carat = 0.2 grams)
# Distribution is Right Skewed
# There are more small diamonds than large
ggplot(diamond_sample, aes(x = carat)) +
  geom_histogram(bins = 40, fill = "steelblue", color = "white", alpha = 0.85) +
  labs(title = "Distribution of Carat", x = "Carat", y = "Frequency") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```

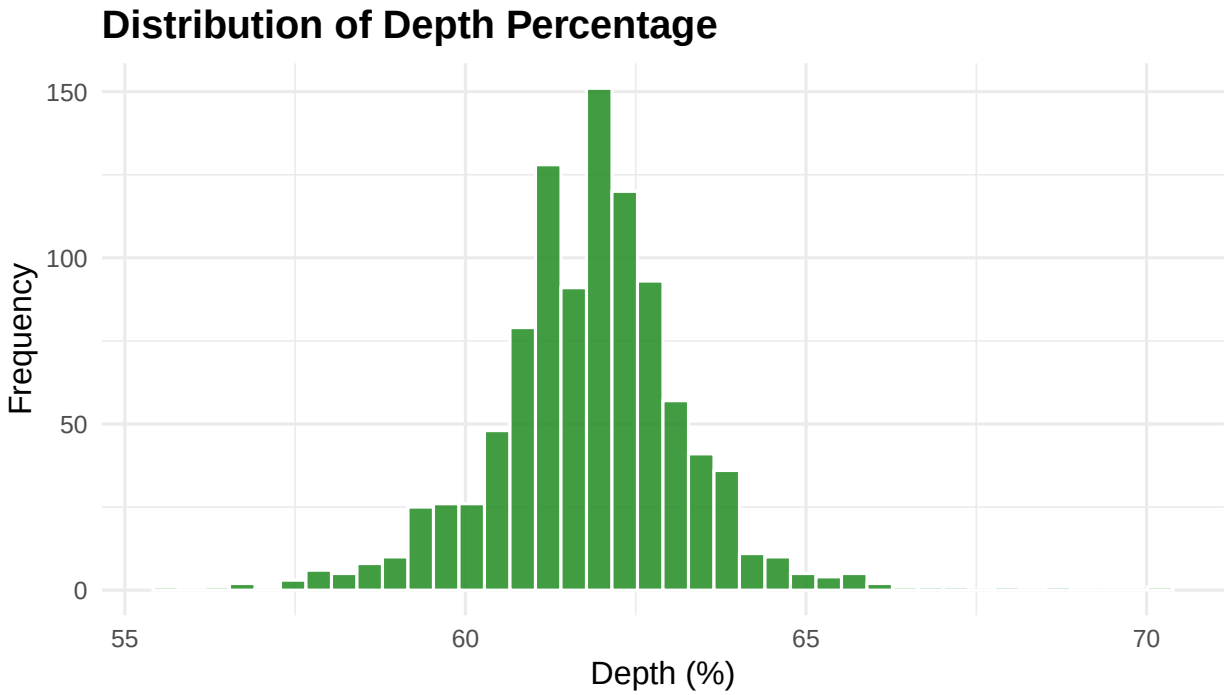



```
# Price - Cost of Diamond (USD)
# Distribution is Right Skewed
```

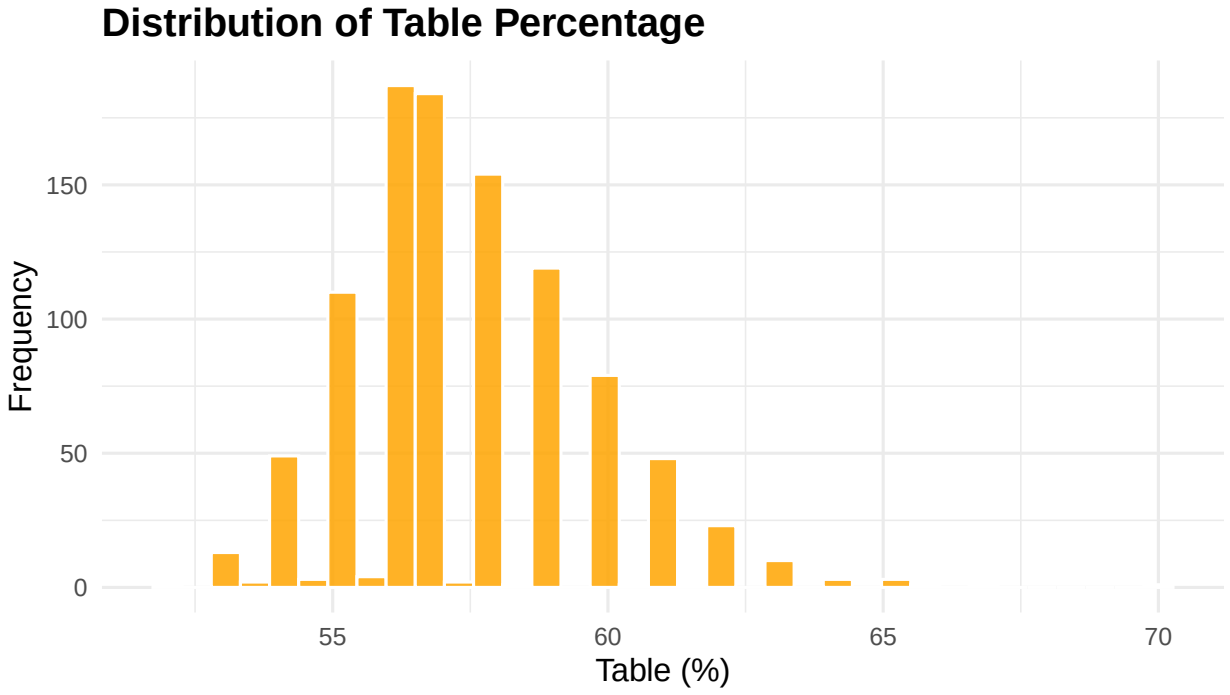
```r
# There are more lower-priced diamonds than expensive
ggplot(diamond_sample, aes(x = price)) +
  geom_histogram(bins = 40, fill = "red", color = "white", alpha = 0.85) +
  labs(title = "Distribution of Price", x = "Price (USD)", y = "Frequency") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```



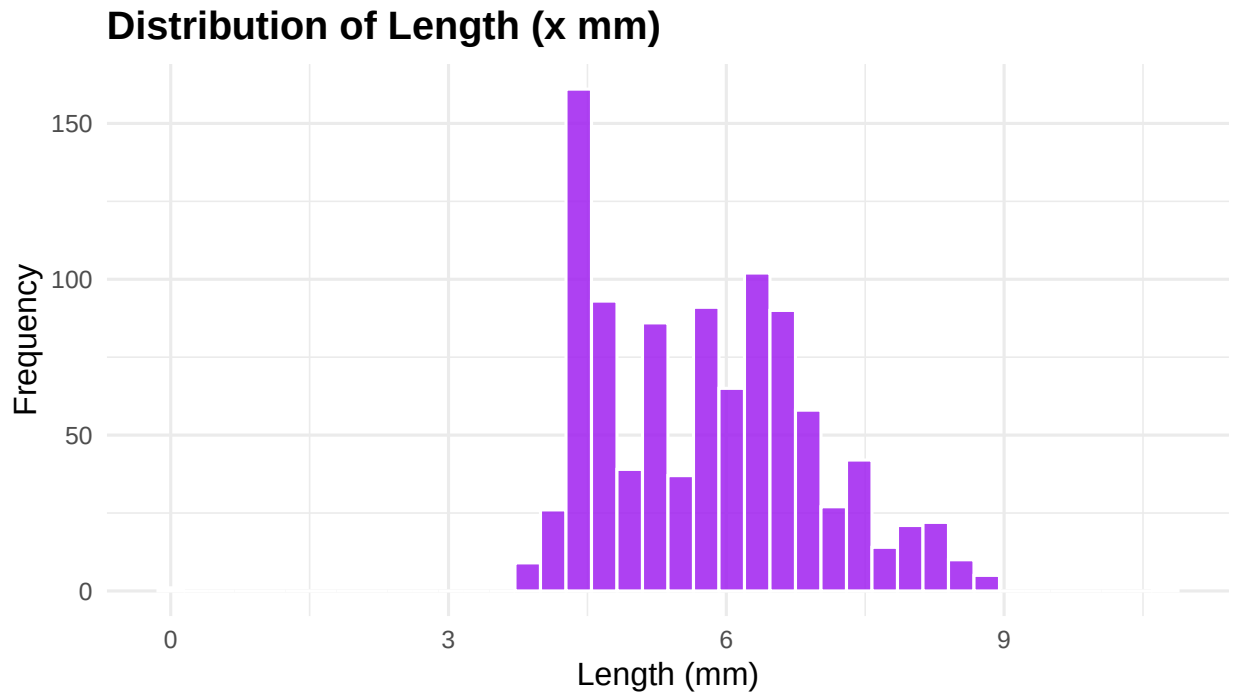

```r
# Depth Percentage - Ratio of the stone's total height to its width
# Distribution is Approximately Normal around ~62%
ggplot(diamond_sample, aes(x = depth)) +
  geom_histogram(bins = 40, fill = "forestgreen", color = "white", alpha = 0.85) +
  labs(title = "Distribution of Depth Percentage", x = "Depth (%)",
       y = "Frequency") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```



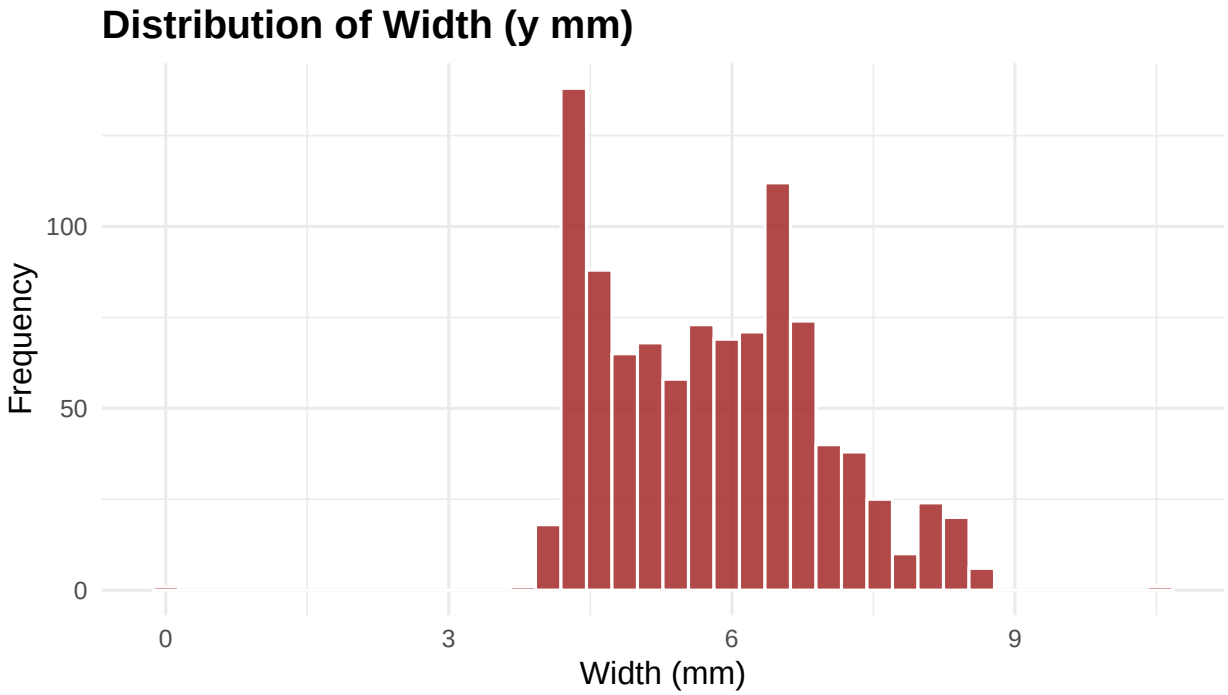

```r
# Table Percentage - Ratio of the width of the stone's flat top facet (table) to its entire width or di
# Distribution is Left Skewed and roughly symmetric
ggplot(diamond_sample, aes(x = table)) +
  geom_histogram(bins = 35, fill = "orange", color = "white", alpha = 0.85) +
  labs(title = "Distribution of Table Percentage", x = "Table (%)",
       y = "Frequency") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```



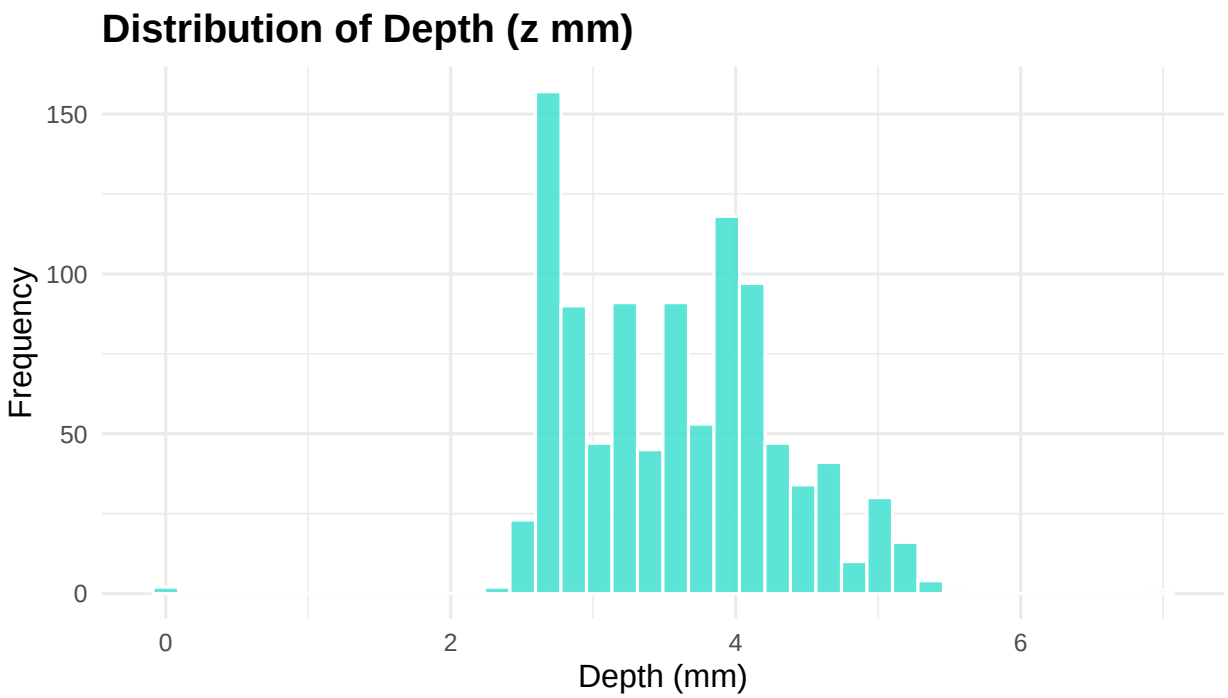
```r
# Length (x) - Length of Diamond in mm
# Distribution is Right Skewed
ggplot(diamond_sample, aes(x = x)) +
  geom_histogram(bins = 40, fill = "purple", color = "white", alpha = 0.85) +
  labs(title = "Distribution of Length (x mm)", x = "Length (mm)",
       y = "Frequency") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```




```r
# Width (y) - Width of Diamond in mm
# Distribution is Right Skewed
ggplot(diamond_sample, aes(x = y)) +
  geom_histogram(bins = 40, fill = "brown", color = "white", alpha = 0.85) +
  labs(title = "Distribution of Width (y mm)", x = "Width (mm)", y = "Frequency") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```



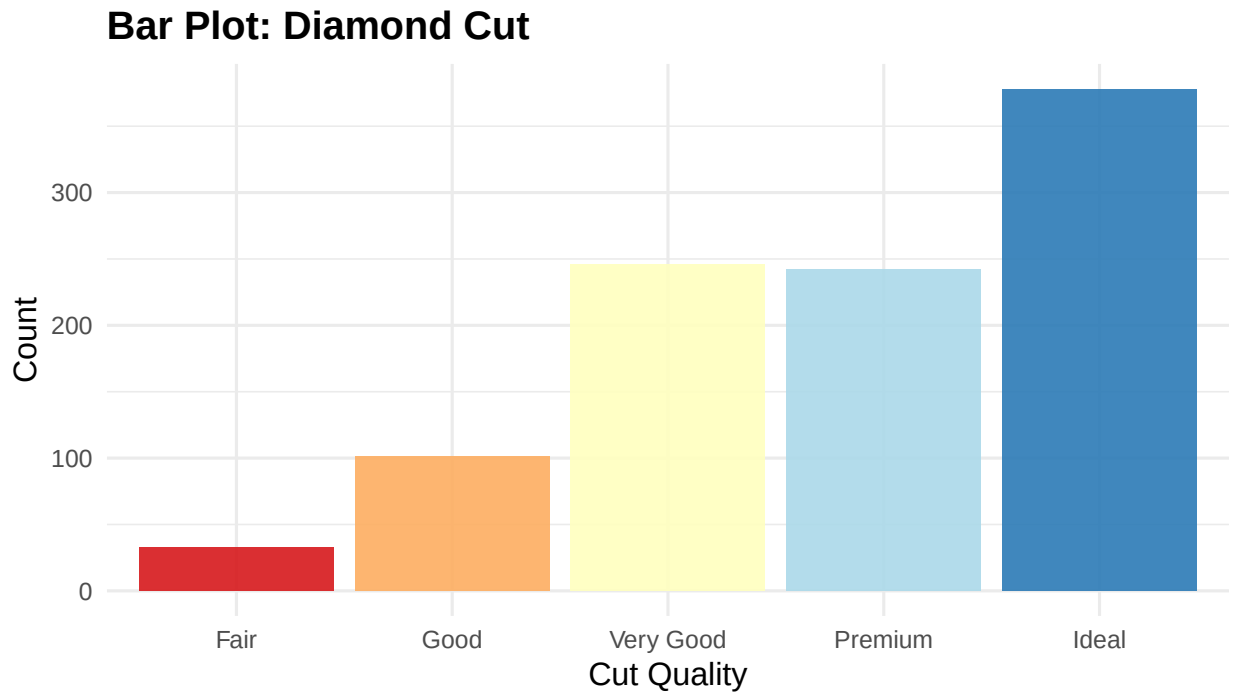

```
# Depth (z) - Depth of Diamond in mm
# Distribution is Right Skewed
ggplot(diamond_sample, aes(x = z)) +
  geom_histogram(bins = 40, fill = "turquoise", color = "white", alpha = 0.85) +
  labs(title = "Distribution of Depth (z mm)", x = "Depth (mm)", y = "Frequency") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```



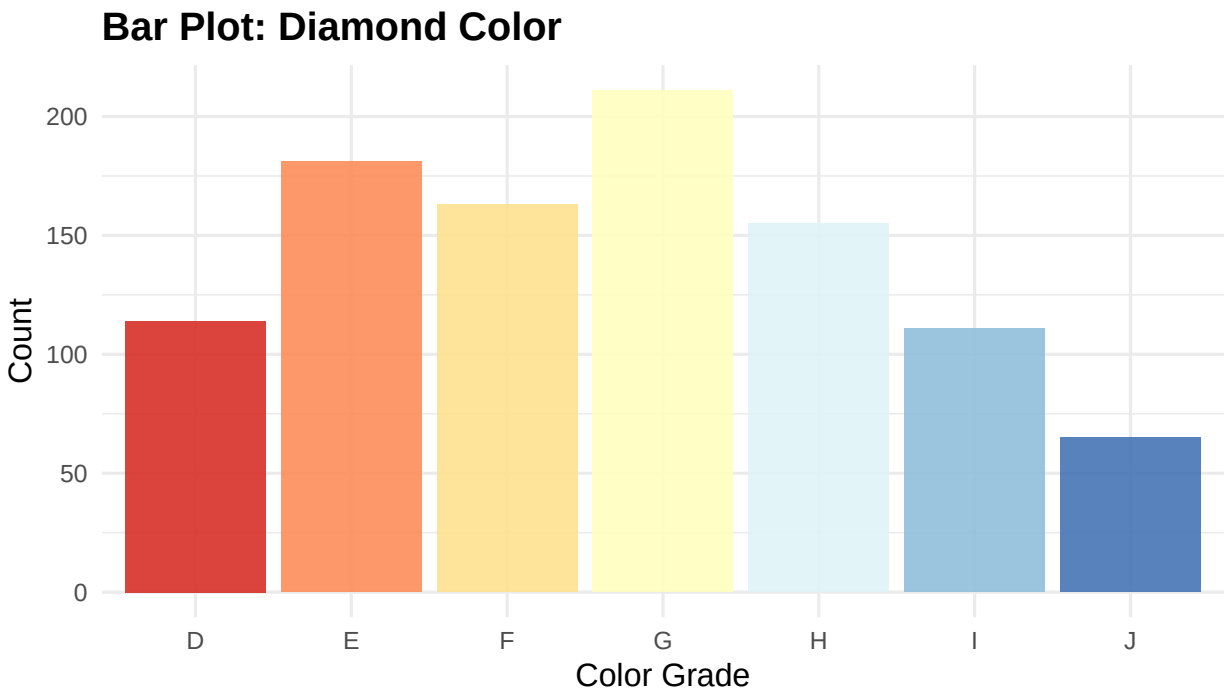
Categorical Variables

```r
# Reordering Categorical Variables
diamond_sample$cut <- factor(diamond_sample$cut,
                             levels = c("Fair", "Good", "Very Good",
                                        "Premium", "Ideal"))
diamond_sample$color <- factor(diamond_sample$color,
                               levels = c("D", "E", "F", "G", "H", "I", "J"))
diamond_sample$clarity <- factor(diamond_sample$clarity,
                                 levels = c("I1", "SI2", "SI1", "VS2",
                                            "VS1", "VVS2", "VVS1", "IF"))

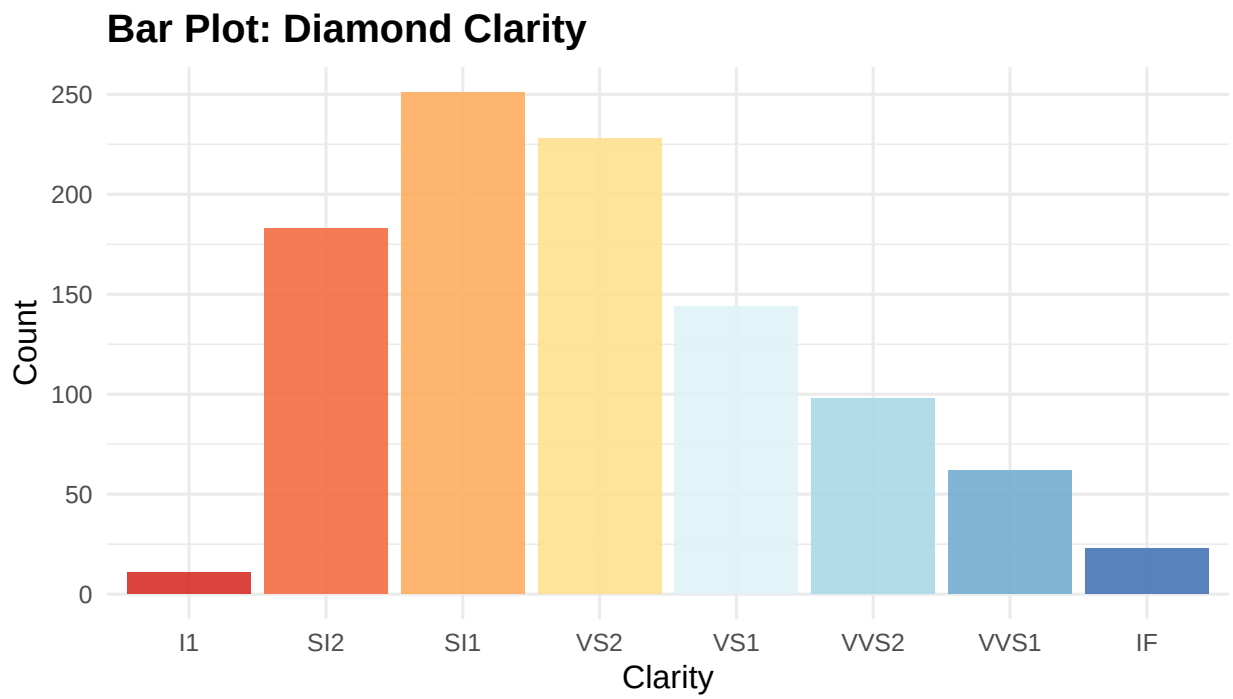

# Cut - Quality of the Diamond's Design
# Stone's with better cut grades reflect more light
# Ideal is the most common grade, fair is the least common
ggplot(diamond_sample, aes(x = cut, fill = cut)) +
  geom_bar(alpha = 0.9, show.legend = FALSE) +
  scale_fill_brewer(palette = "RdYlBu") +
  labs(title = "Bar Plot: Diamond Cut",
       x = "Cut Quality", y = "Count") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```




```r
# Color - Absense of Color in Diamond
# D, E, and F considered colorless (D having the greatest absense of color)
# G, H, I, J considered near colorless (J having the least absense of color)
# Mid-range grades (E-H) most common
ggplot(diamond_sample, aes(x = color, fill = color)) +
  geom_bar(alpha = 0.9, show.legend = FALSE) +
  scale_fill_brewer(palette = "RdYlBu") +
  labs(title = "Bar Plot: Diamond Color",
       x = "Color Grade", y = "Count") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```




```
# Clarity - Measurement of Inclusions and Blemishes
# IF - Internally Flawless
# VVS1 & VVS2 - Very, Very Slightly Included
# VS1 & VS2 - Very Slightly Included
# SI1 & SI2 - Slightly Included
# I1 - Included
# SI1 and VS2 are most common; IF is least common
ggplot(diamond_sample, aes(x = clarity, fill = clarity)) +
  geom_bar(alpha = 0.9, show.legend = FALSE) +
  scale_fill_brewer(palette = "RdYlBu") +
  labs(title = "Bar Plot: Diamond Clarity",
       x = "Clarity", y = "Count") +
  theme_minimal(base_size = 13) + theme(plot.title = element_text(face = "bold"))
```



3) Correlation Between Selected Variables

Quantitative: Carat, Table, Length (x) | **Categorical:** Cut, Clarity

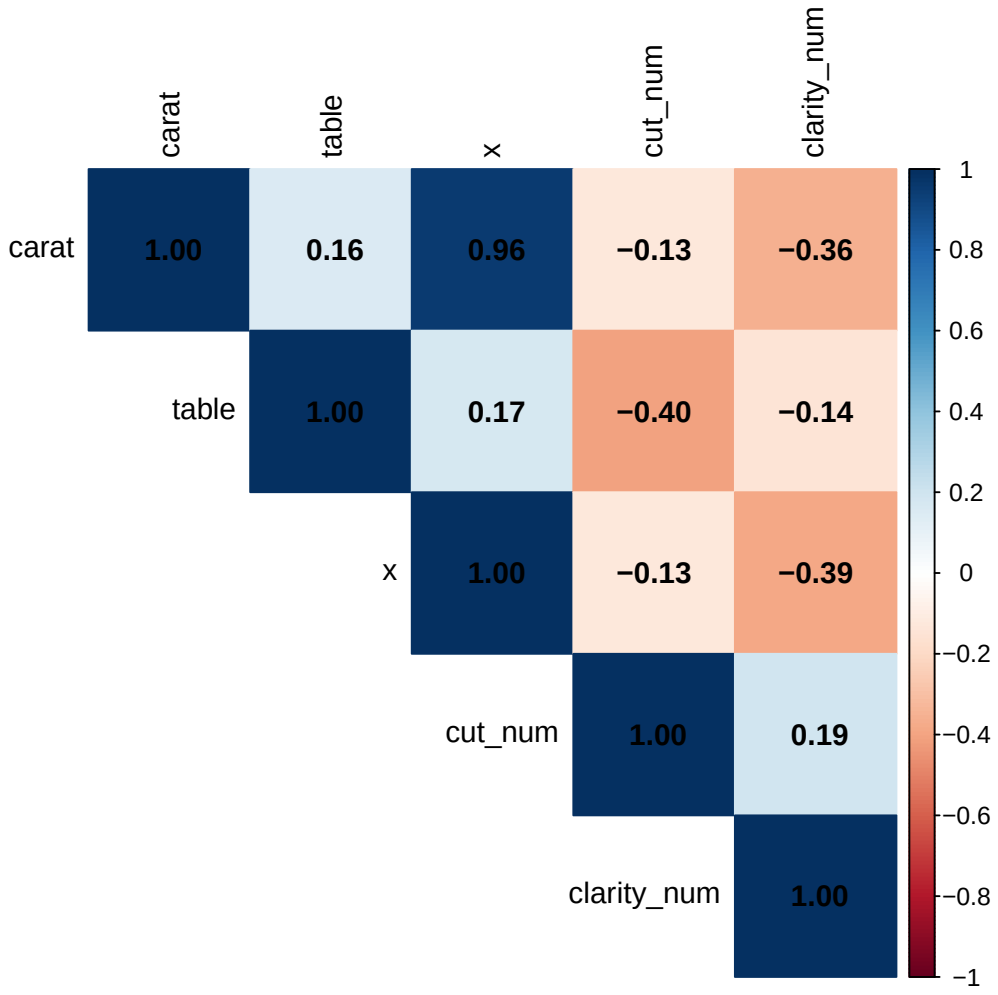
```
cor_data <- diamond_sample %>%
  mutate(cut_num   = as.numeric(cut),     # 1=Fair ... 5=Ideal
         clarity_num = as.numeric(clarity))  # 1=I1 ... 8=IF

cor_vars <- cor_data[ , c("carat", "table", "x", "cut_num", "clarity_num")]
cor_matrix <- cor(cor_vars, use = "complete.obs")

# Correlation Matrix
print(cor_matrix)
```

```
##                  carat      table          x    cut_num clarity_num
## carat        1.0000000  0.1583170  0.9556272 -0.1269788  -0.3572331
## table        0.1583170  1.0000000  0.1742509 -0.4012575  -0.1401418
## x            0.9556272  0.1742509  1.0000000 -0.1264365  -0.3864220
## cut_num     -0.1269788 -0.4012575 -0.1264365  1.0000000   0.1930878
## clarity_num -0.3572331 -0.1401418 -0.3864220  0.1930878   1.0000000
```

```
corrplot(cor_matrix,
         method   = "color",
         type     = "upper",
         addCoef.col = "black",
         tl.col   = "black",
         title    = "Correlation Matrix: Selected Variables",
         mar      = c(0, 0, 2, 0))
```



- **Strong positive** correlation between length (x) and carat.
- **Weak positive** correlation between cut and clarity, carat and table, and x and table.
- **Weak negative** correlation between clarity and table, cut and carat, cut and x.
- **Moderate negative** correlation between clarity and carat, and clarity and x.

4) Multiple Linear Regression Model

```
# Model using previously chosen variables -
# Carat, Table, Length (x), Cut, Clarity
model_1 <- lm(price ~ carat + table + x + cut + clarity,
              data = diamond_sample)
summary(model_1)
```

```
##
## Call:
## lm(formula = price ~ carat + table + x + cut + clarity, data = diamond_sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16373.6   -600.9   -128.5    388.7   7054.3
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -5125.95    1468.42  -3.491 0.000503 ***
## carat         9217.22     284.05  32.449  < 2e-16 ***
## table          -34.27      22.86  -1.499 0.134193
## x             -431.92     125.78  -3.434 0.000620 ***
## cutGood       1094.57     267.41   4.093 4.60e-05 ***
## cutVery Good  1650.58     249.05   6.628 5.61e-11 ***
## cutPremium    1625.82     248.06   6.554 9.01e-11 ***
## cutIdeal      1734.40     252.88   6.859 1.23e-11 ***
## claritySI2    3543.25     416.58   8.506  < 2e-16 ***
## claritySI1    4124.99     416.06   9.914  < 2e-16 ***
## clarityVS2    4755.07     416.48  11.417  < 2e-16 ***
## clarityVS1    5204.87     423.64  12.286  < 2e-16 ***
## clarityVVS2   5590.63     430.81  12.977  < 2e-16 ***
## clarityVVS1   5389.19     444.60  12.121  < 2e-16 ***
## clarityIF     5244.27     497.42  10.543  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1326 on 985 degrees of freedom
## Multiple R-squared:  0.9035, Adjusted R-squared:  0.9021
## F-statistic: 658.8 on 14 and 985 DF,  p-value: < 2.2e-16
```

5) Comments

For the frequency data, I correctly expected to see that larger, higher quality diamonds were more rare than smaller, lower quality ones. Since high quality diamonds have higher value partly due to their rarity, it makes sense that they would be less common in the sample.

I was intrigued by the similarity in histograms for length (x), width (y), and depth (z). While it makes sense the three variables would share a similar distribution, I was surprised to find their right skewed distributions as nearly identical.

One result I anticipated was a strong correlation between Carat and length (x), which I expect remains strong for Carat and the other measurements of width (y) and depth (z). Intuitively, it makes sense that as the diamond gets bigger in size, its weight also increases.

Another point of interest was the weak to moderate negative correlation between the chosen categorical variables of cut and clarity, to the quantitative variables. This implies that as cut and clarity conditions improve, size and weight tend to decrease.

PART 2 — Simple Linear Regression

1) Simple Linear Regression Analysis

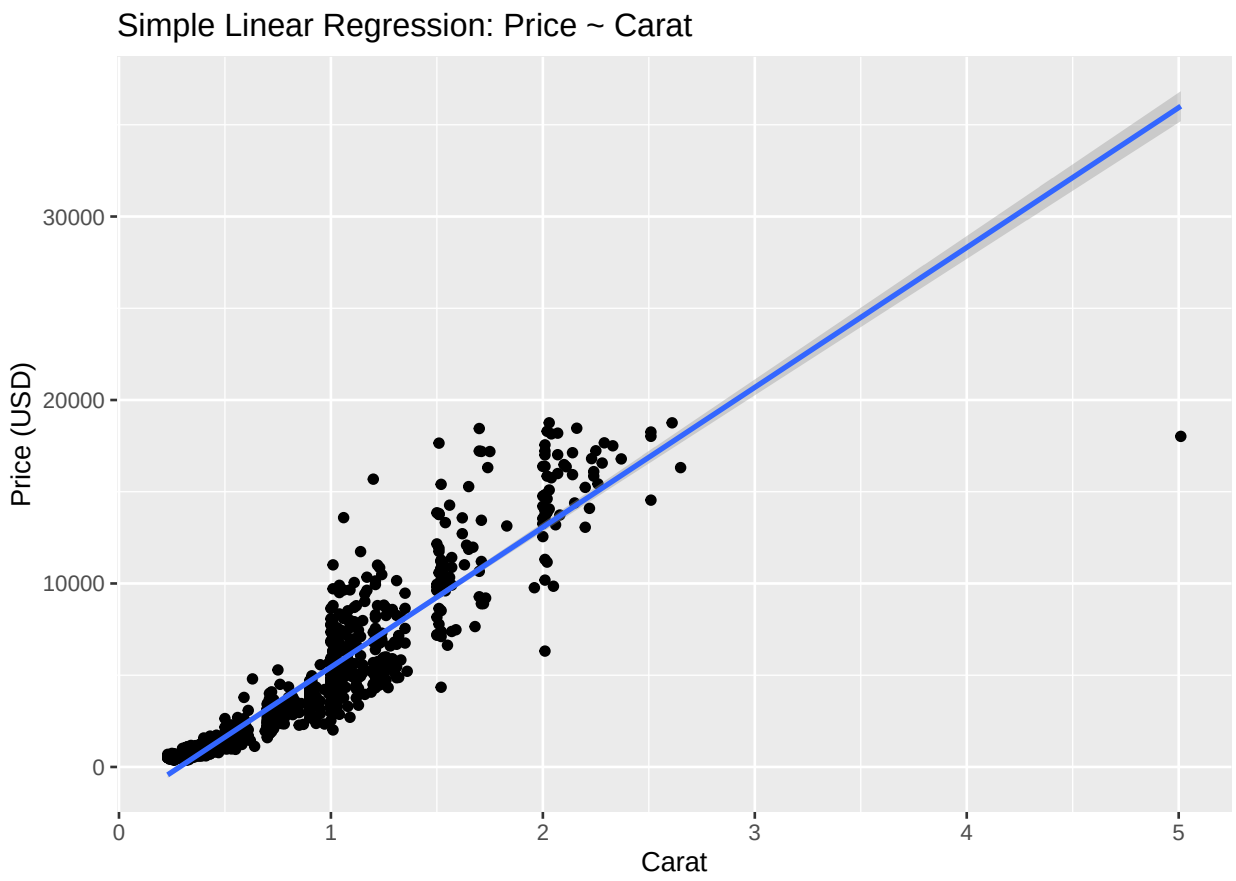
Simple Linear Regression Analysis

```r
# Simple Model using Carat and Price
simple_model <- lm(price ~ carat, data = diamond_sample)
simple_model
```

```
##
## Call:
## lm(formula = price ~ carat, data = diamond_sample)
```

```
##
## Coefficients:
## (Intercept)        carat
##       -2182         7623
```

```
ggplot(diamond_sample, aes(x = carat, y = price)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = T) +
  labs(title    = "Simple Linear Regression: Price ~ Carat",
       x = "Carat", y = "Price (USD)")
```



2) Summary Statistics

```
# Summary Statistics
summary(simple_model)
```

```
##
## Call:
## lm(formula = price ~ carat, data = diamond_sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17993.5   -790.0    -12.6    570.1   8719.5
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -2181.53      97.18  -22.45   <2e-16 ***
## carat        7623.36      99.08   76.94   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1611 on 998 degrees of freedom
## Multiple R-squared:  0.8557, Adjusted R-squared:  0.8556
## F-statistic:  5920 on 1 and 998 DF,  p-value: < 2.2e-16
```

```
# Confidence Interval for Coefficients
confint(simple_model, level = 0.95)
```

```
##                 2.5 %    97.5 %
## (Intercept) -2372.229 -1990.836
## carat        7428.929  7817.800
```

```
# For a new observation where Carat = 1
new_obs <- data.frame(carat = 1.0)

# Confidence Interval
predict(simple_model, newdata = new_obs, interval = "confidence", level = 0.95)
```

```
##        fit      lwr      upr
## 1 5441.832 5336.878 5546.786
```

```
# Prediction Interval
predict(simple_model, newdata = new_obs, interval = "prediction", level = 0.95)
```

```
##        fit      lwr      upr
## 1 5441.832 2279.482 8604.182
```

Coefficients (Estimate)

- **Intercept: $-2181.53** | When carat is 0, the estimated price is $-2181.53. This estimate is not meant for practical use, rather it serves as a baseline for the model. Std. Error is 97.18 and t value is -22.45. These indicate uncertainty around the intercept estimate.
- **Carat: $7623.36** | For every increase in carat amount by 1, price increases by $7623.36 on average. Std. Error is 99.08 and t value is 76.94. These indicate uncertainty around the carat estimate.

Hypothesis Tests

- Intercept | $H_0 : \beta_0 = 0; H_1 : \beta_0 \neq 0$ | p-value =<2e-16 | Reject H_0 at $\alpha = 0.05$, Intercept is significant
- Carat (Slope) | $H_0 : \beta_1 = 0; H_1 : \beta_1 \neq 0$ | p-value = <2e-16 | Reject H_0 at $\alpha = 0.05$, Carat is a significant predictor

Residual Standard Error

- Residual standard error: 1611 on 998 degrees of freedom
- The RSE is the typical deviation of observed price values from the model's predicted price values. The value of 1611 indicates that on average the predicted price will deviate from the actual price by 1611.

R^2 and AdjustedR^2

- Multiple R^2: 0.8557 | Approximately 85.57% of variation in price is explained by the model
- Adjusted R^2: 0.8556 | When penalizing for the number of predictors, the percentage remains highly similar at 85.56% since there is only one predictor

F-Statistic

- F-statistic: 5920 on 1 and 998 DF | p-value: $<$ 2.2e-16
- Since p-value $< \alpha = 0.05$, the model overall is significant

Coefficient Confidence Intervals

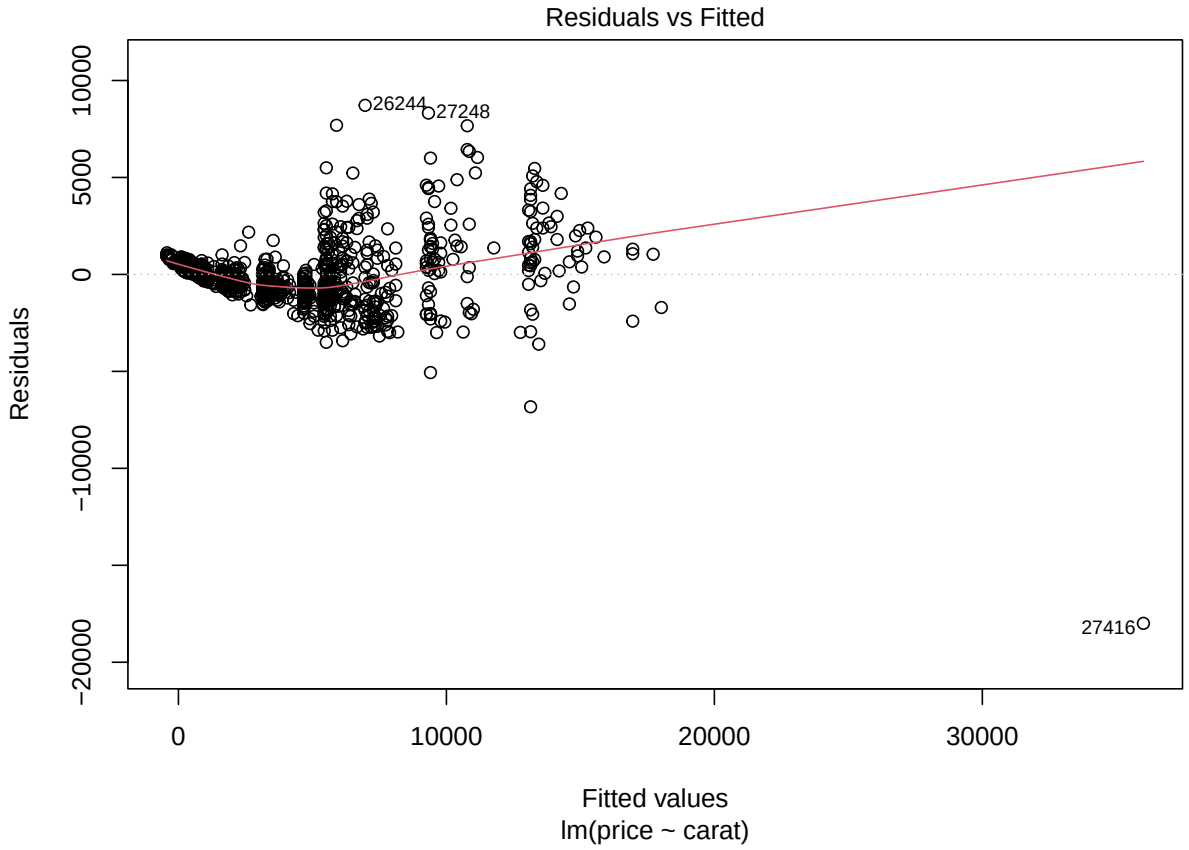
- 95% confident that the true intercept is between $-2,372.229 and -$1,990.836
- 95% confident that each additional carat increases price by between $7,428.929 and $7,817.8

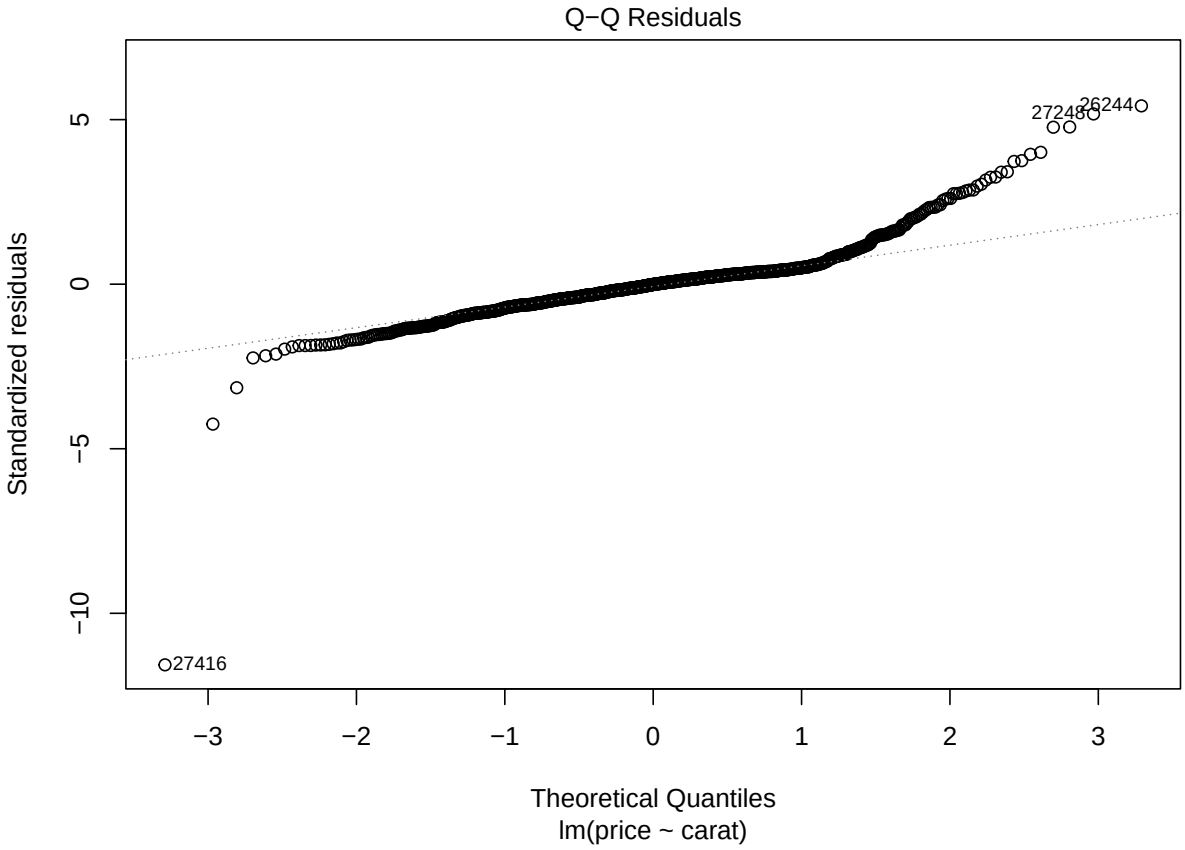
Interval Estimates when Carat = 1

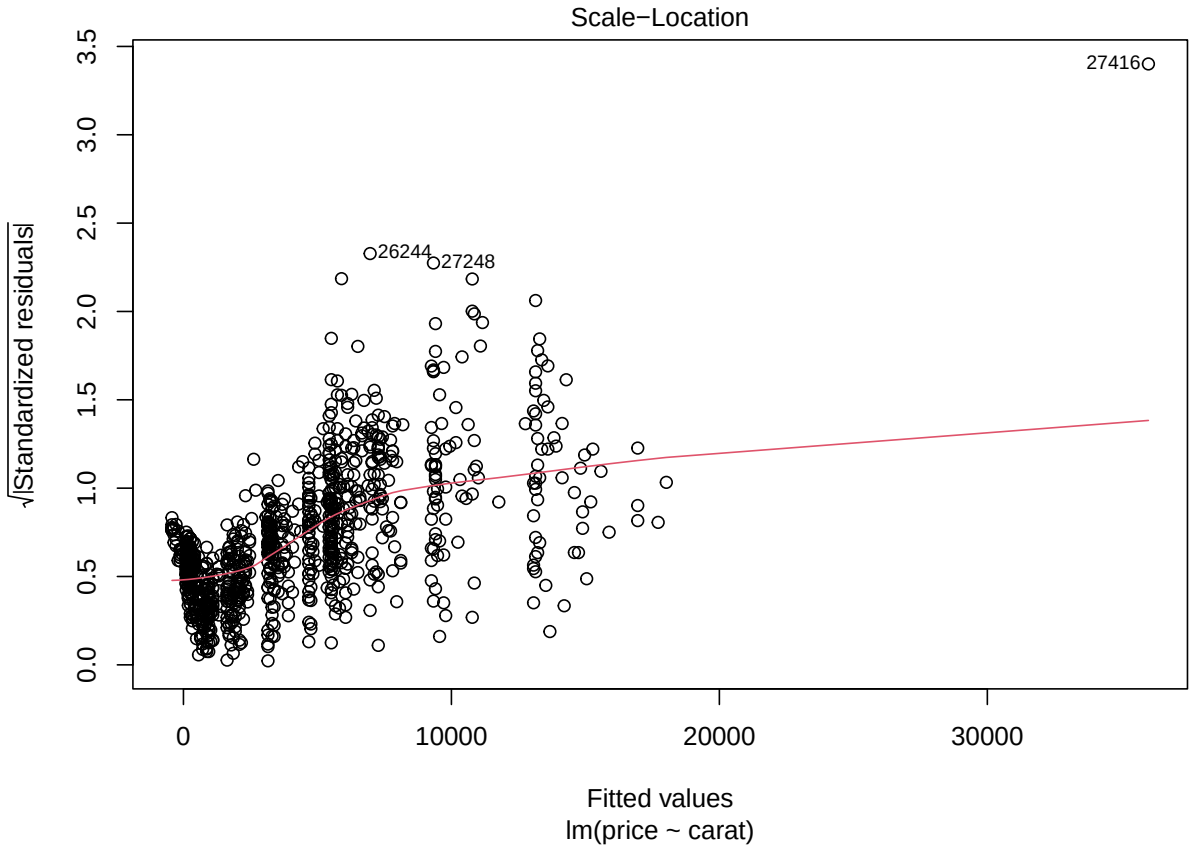
- **Confidence Interval:** $5,336.878 to $5,546.786 | 95% confident that for a 1-Carat diamond, the true mean price falls between $5,336.878 and $5,546.786
- **Prediction Interval:** $2,279.482 to $8,604.182 | 95% confident that for one specific 1-Carat diamond, the true price falls between $2,279.482 and $8,604.182

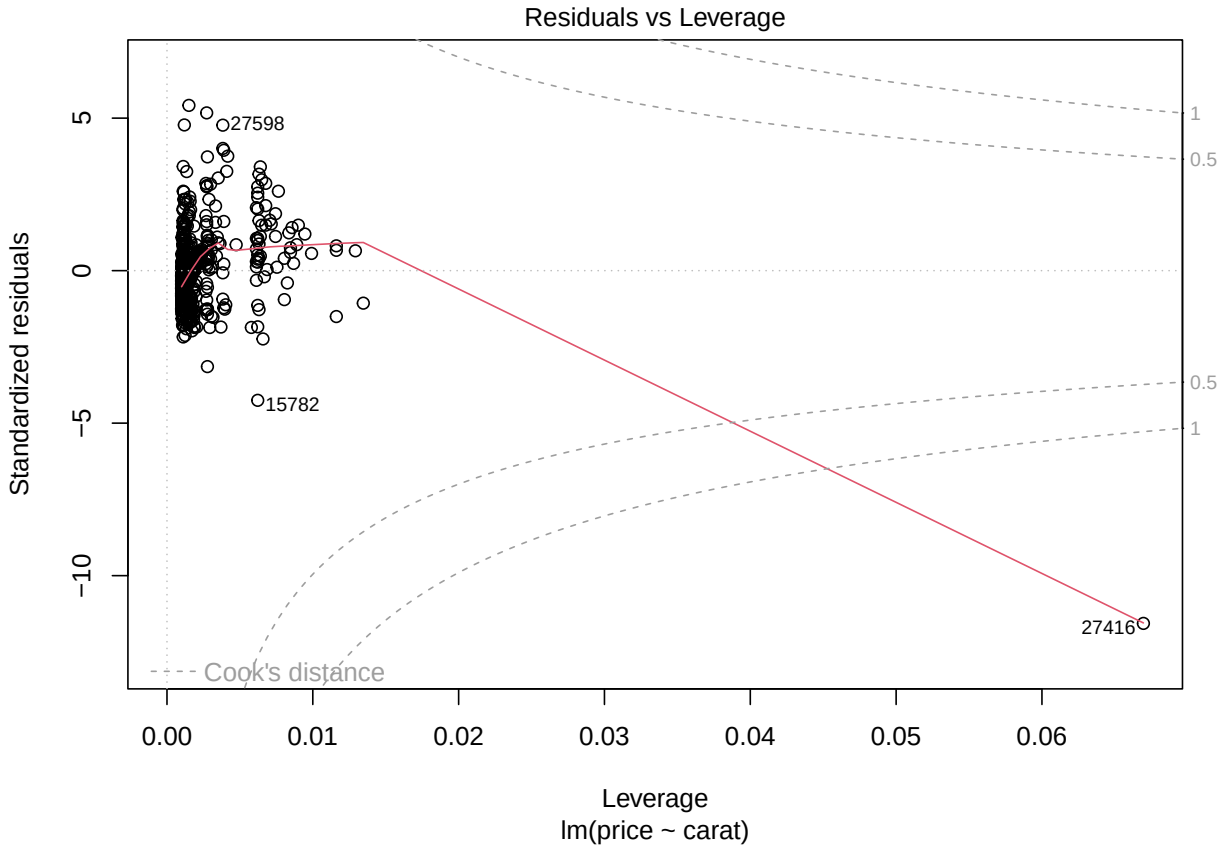
3) Assumption Testing and Transformation

```
# Test Assumptions
plot(simple_model)
```




Q−Q Residuals
Standardized residuals
5
0
−5
−10
27248
26244
27416
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(price ~ carat)


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
26244
27248
27416
0
10000
20000
30000
Fitted values
lm(price ~ carat)

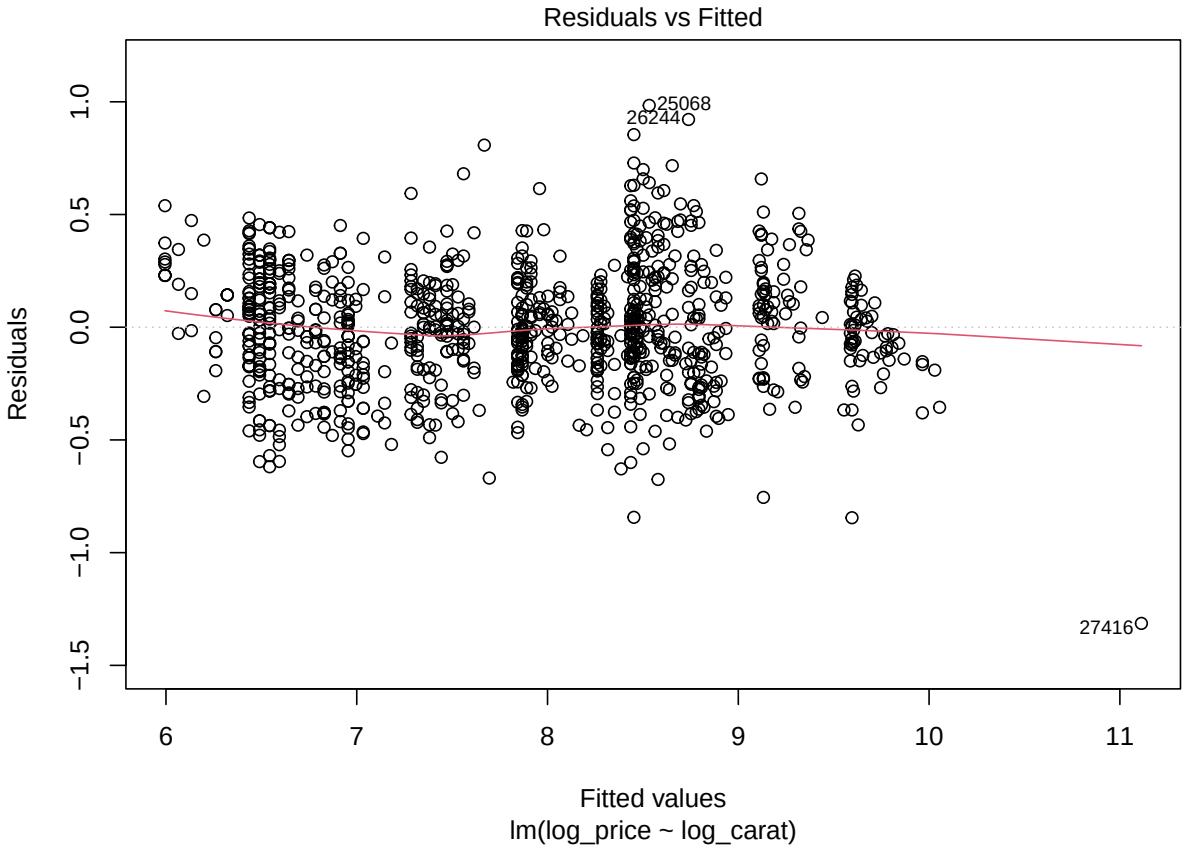


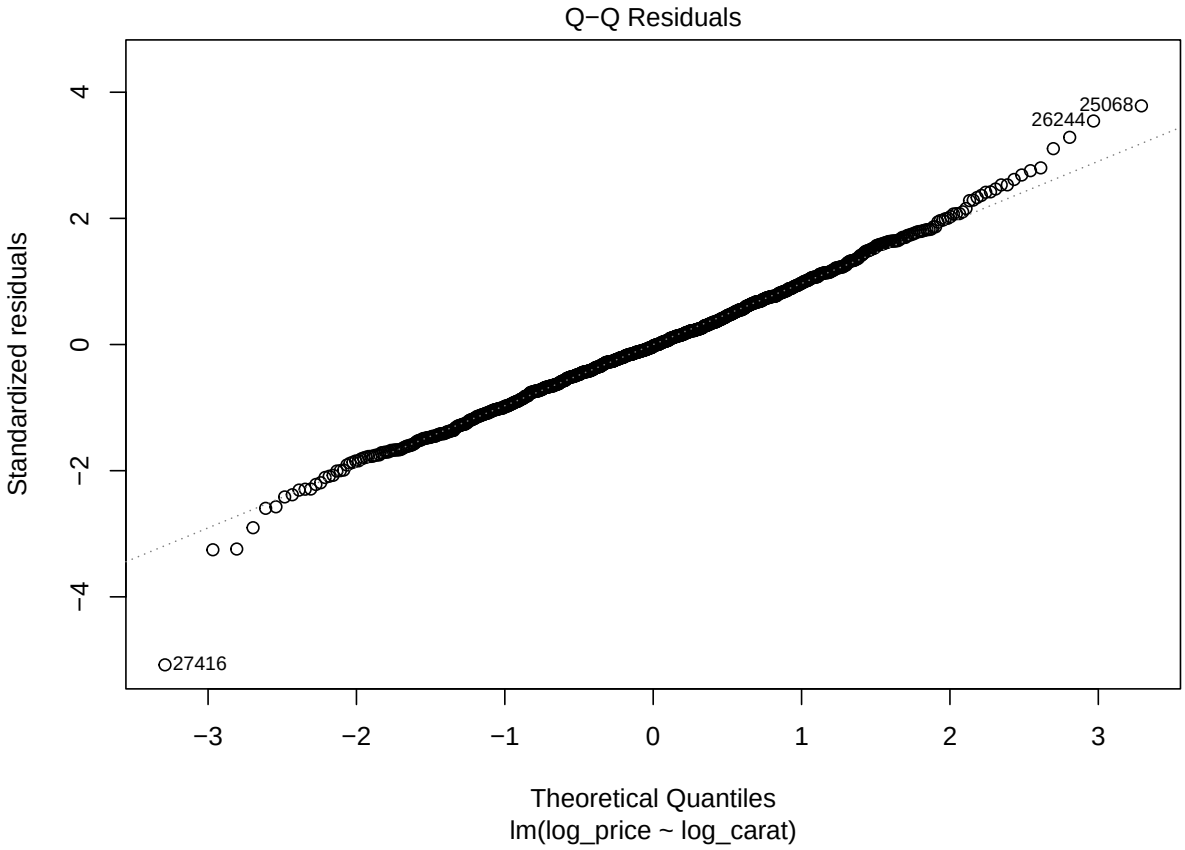
- **Residuals vs Fitted:** Variance increases as fitted values increase > heteroscedasticity violated.
- **Normal Q-Q:** Heavy upper tail > normality of residuals violated.
- **Scale-Location:** Variance increases as the value of fitted values increases > heteroscedasticity violated
- **Residuals vs Leverage:** One outlier identified outside Cook's distance line.

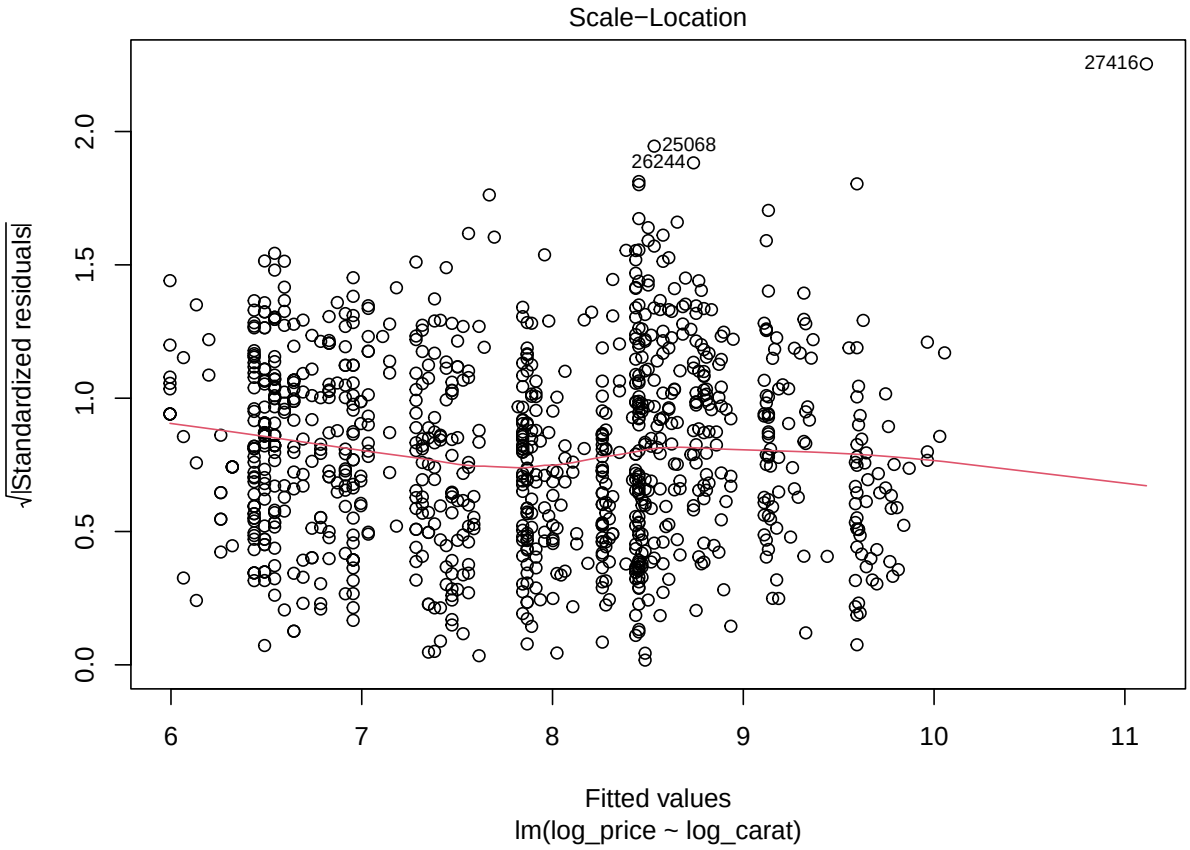
```
# Log Transformation
diamond_sample_log <- diamond_sample %>%
  mutate(log_price = log(price),
         log_carat = log(carat))

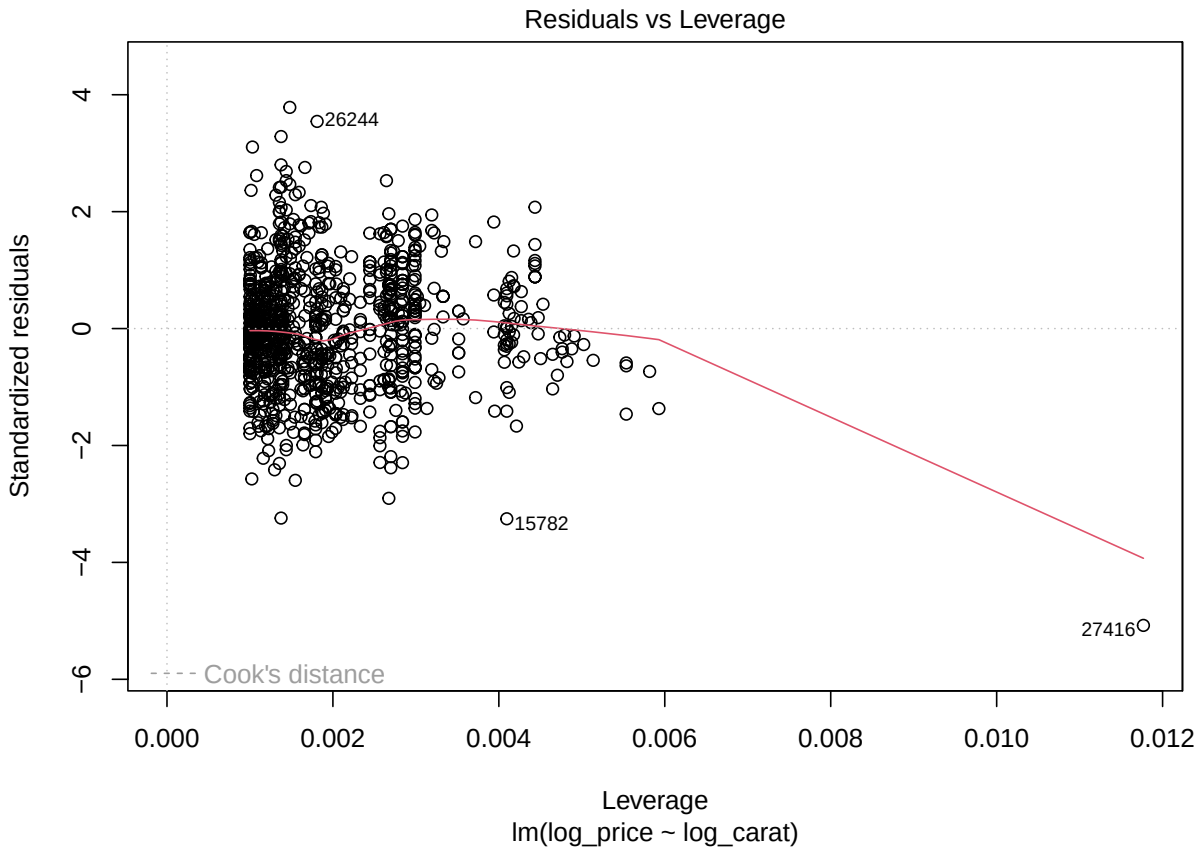
model_log <- lm(log_price ~ log_carat, data = diamond_sample_log)

# Test Assumptions on Log Transformation
plot(model_log)
```

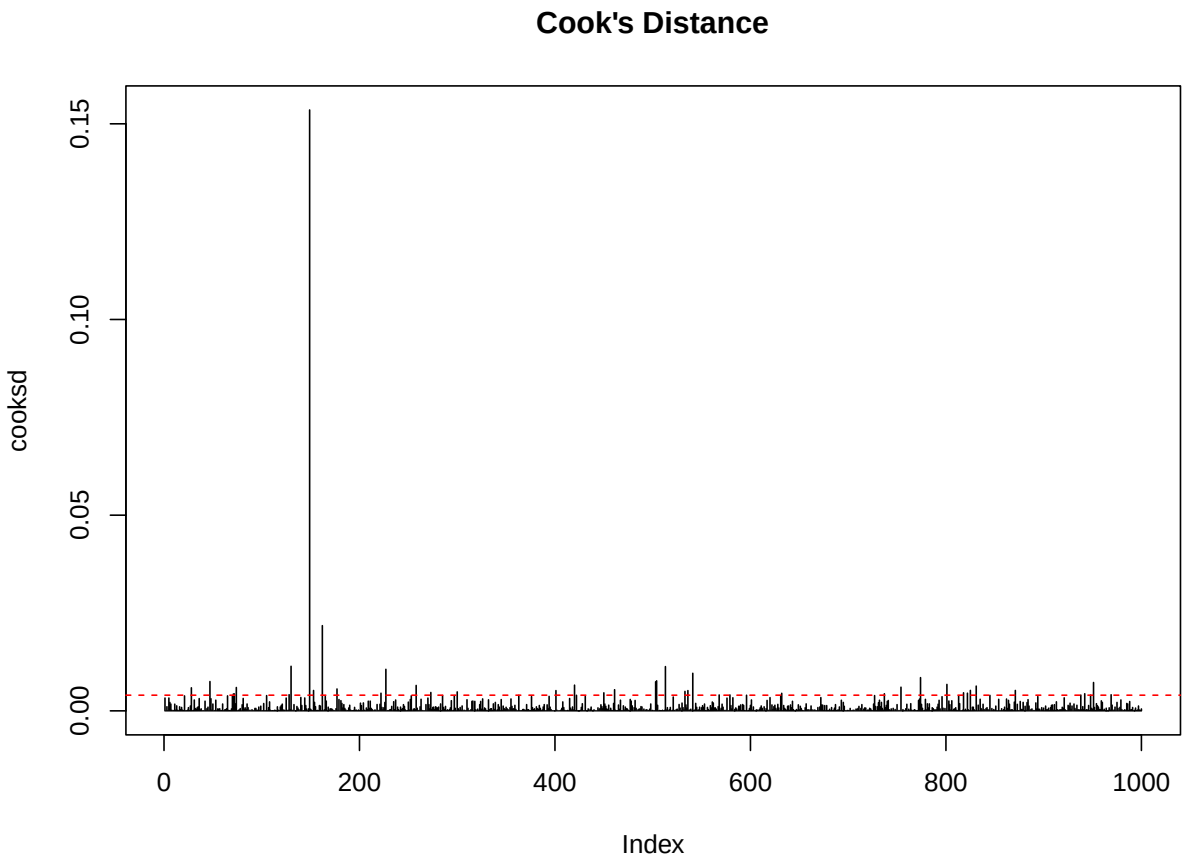

Residuals vs Fitted
25068
26244
27416
Residuals
1.0
0.5
0.0
−0.5
−1.0
−1.5
6
7
8
9
10
11
Fitted values
lm(log_price ~ log_carat)


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(log_price ~ log_carat)
25068
26244
27416
4
2
0
−2
−4
−3
−2
−1
0
1
2
3


Scale−Location
27416
25068
26244
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
6
7
8
9
10
11
Fitted values
lm(log_price ~ log_carat)




```r
# Investigate Outlier
cooksd <- cooks.distance(model_log)
plot(cooksd, type = "h", main = "Cook's Distance")
abline(h = 4/nrow(diamond_sample), col = "red", lty = 2)
```




```
influential <- which(cooksd > 4/nrow(diamond_sample))
print(diamond_sample[influential, ])
```

```
##           X carat       cut color clarity depth table price     x     y    z
## 21784 21784  2.05   Premium     I     SI2  62.0  58.0  9850  8.13  8.08 5.02
## 28278 28278  0.31 Very Good     J     SI1  61.9  59.0   363  4.28  4.32 2.66
## 27075 27075  1.71   Premium     G     VS1  62.3  58.0 17197  7.65  7.61 4.75
## 21987 21987  1.11     Ideal     G      IF  61.2  54.0 10053  6.71  6.73 4.11
## 25649 25649  2.51   Premium     H     SI2  59.3  61.0 14543  8.81  8.77 5.21
## 30249 30249  0.26   Premium     E    VVS2  61.7  61.0   724  4.12  4.08 2.53
## 26244 26244  1.20   Premium     D    VVS1  62.1  59.0 15686  0.00  0.00 0.00
## 27416 27416  5.01      Fair     J      I1  65.5  59.0 18018 10.74 10.54 6.98
## 21842 21842  1.04     Ideal     E    VVS2  61.0  57.0  9901  6.61  6.57 4.02
## 15782 15782  2.01      Fair     G      I1  70.2  57.0  6315  7.53  7.50 5.27
## 26593 26593  2.65   Premium     G     SI2  62.0  61.0 16314  8.82  8.75 5.45
## 21545 21545  1.06     Ideal     G    VVS2  61.4  56.0  9645  6.56  6.62 4.04
## 25068 25068  1.06     Ideal     D    VVS2  61.8  56.0 13588  6.49  6.53 4.03
## 41608 41608  0.32   Premium     J     VS2  61.9  58.0   393  4.35  4.38 2.70
## 44002 44002  0.30      Good     G     SI2  63.3  57.0   394  4.25  4.28 2.70
## 3382   3382  0.31   Premium     J     VS1  60.7  60.0   408  4.34  4.39 2.65
## 26126 26126  1.52   Premium     G    VVS2  62.9  59.0 15404  7.29  7.25 4.57
## 30762 30762  0.25   Premium     E    VVS1  60.7  60.0   740  4.11  4.09 2.49
## 8054   8054  0.23 Very Good     F    VVS2  61.6  57.0   583  3.96  3.99 2.45
## 21622 21622  1.01   Premium     E    VVS2  62.4  58.0  9711  6.42  6.36 3.99
```

```
## 22967 22967  1.01     Ideal     D    VVS2  61.5  57.0 11015  6.43  6.48 3.97
## 34929 34929  0.32      Good     J     SI1  63.2  56.0   374  4.31  4.36 2.74
## 8101   8101  1.52   Premium     G      I1  62.3  60.0  4345  7.34  7.28 4.56
## 10512 10512  0.63 Very Good     D      IF  59.6  59.0  4800  5.55  5.59 3.32
## 28641 28641  0.33     Ideal     G     SI2  61.9  56.0   434  4.41  4.45 2.74
## 29079 29079  0.23   Premium     D    VVS2  58.9  60.0   688  4.03  3.95 2.35
## 21549 21549  1.09     Ideal     D     VS1  62.3  56.0  9650  6.63  6.59 4.12
## 36832 36832  0.30     Ideal     E      IF  61.1  57.0   957  4.31  4.33 2.64
## 31288 31288  0.33     Ideal     F     SI2  62.3  55.0   450  4.44  4.49 2.78
## 10044 10044  0.31     Ideal     J     SI1  61.3  56.0   418  4.37  4.40 2.69
## 27598 27598  1.70     Ideal     G    VVS1  61.0  56.0 18445  7.67  7.62 4.66
## 27248 27248  1.51     Ideal     E     VS1  62.2  57.0 17649  7.37  7.32 4.57
## 40       40  0.33     Ideal     I     SI2  61.8  55.0   403  4.49  4.51 2.78
## 21391 21391  1.04   Premium     F    VVS2  61.6  59.0  9505  6.53  6.49 4.01
## 27087 27087  1.70 Very Good     G     VS1  61.8  53.0 17219  7.63  7.67 4.73
## 53666 53666  1.09   Premium     I     SI2  62.3  62.0  2708  6.63  6.57 4.11
## 23682 23682  1.14 Very Good     F    VVS1  62.2  55.9 11737  6.67  6.69 4.16
## 38075 38075  0.30     Ideal     E    VVS2  61.7  57.0  1013  4.34  4.31 2.67
## 20456 20456  1.01 Very Good     E    VVS2  63.4  57.0  8797  6.45  6.33 4.05
## 38604 38604  0.31 Very Good     E    VVS1  60.0  56.0  1039  4.45  4.49 2.68
## 22955 22955  1.22     Ideal     G    VVS2  60.0  58.0 11003  6.95  6.99 4.18
## 48765 48765  1.01   Premium     I      I1  59.2  57.0  2017  6.58  6.44 3.85
## 22149 22149  2.01      Good     I     SI1  64.1  60.0 10184  7.92  7.88 5.06
```

Assumptions appear valid when analyzing residual plots. However, one outlier appears and should be further investigated. Row 26244 has x, y, and z values of 0, indicating a data entry error. This entry should be removed from the diamond sample.

```
# Remove Entry from diamond sample
diamond_sample_log <- diamond_sample_log %>%
  filter(!(x == 0 | y == 0 | z == 0))

# Refit the Model
model_log <- lm(log_price ~ log_carat, data = diamond_sample_log)
```

4) Summary of Transformed Model

```
summary(model_log)
```

```
##
## Call:
## lm(formula = log_price ~ log_carat, data = diamond_sample_log)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.31036 -0.16950 -0.00554  0.17062  0.98547
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 8.434751   0.009546   883.6   <2e-16 ***
## log_carat   1.659847   0.013654   121.6   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2589 on 996 degrees of freedom
```

```
## Multiple R-squared:  0.9369, Adjusted R-squared:  0.9368
## F-statistic: 1.478e+04 on 1 and 996 DF,  p-value: < 2.2e-16
```

Notable Changes

- Multiple R^2 increases from 0.8557 to 0.9368, indicating the log model captures a greater proportion of variance in price than the simple model
- The estimates changed to 8.434751 for the intercept and 1.659847 for the carat predictor. This represents that for every 1% increase in carat, there is a 1.66% increase in price.

5) Adding Variables to Improve the Model

```
# Final model after adding variables to maximize Adjusted R^2
model_log_R2 <- lm(log_price ~ log_carat + cut + clarity + color,
                   data = diamond_sample_log)

# Adjusted R Squared
summary(model_log_R2)$adj.r.squared
```

```
## [1] 0.9838593
```

Final Model: `log_price ~ log_carat + cut + clarity + color` | Adjusted $R^2 = 0.9839$

6) Comments

One result that surprised me was the extreme outlier from row 26422 which had x, y, and z values of 0. I completely missed this outlier from before, not noticing the tiny bar at 0 for their respective histograms in part 1. Removing this outlier understandably led to an increase in adjusted R squared, and will improve the model.

Another result that intrigued me was the dramatic difference between the confidence and prediction intervals for 1-Carat diamonds in the simple model. While I anticipated the prediction interval to be greater than the confidence interval, I did not expect the difference to be by over $3,000, especially when the confidence interval was so narrow.

I also found it interesting that the final model when intended to maximize adjusted R squared only utilizes one quantitative variable. While I understand that most quantitative variables measure some aspect of size, which is correlated with weight, I anticipated that the model would use at least one other variable in addition to carat.

PART 3 — Model Selection, Multicollinearity, and Prediction

1) Best Model Using AIC

```
# Full Model with log of price and carat
model_full <- lm(log_price ~ log_carat + cut + clarity + color +
                   depth + table + x + y + z, data = diamond_sample_log)

model_aic <- stepAIC(model_full, direction = "both", trace = FALSE)
```

```
summary(model_aic)
```

```
##
## Call:
## lm(formula = log_price ~ log_carat + cut + clarity + color +
##     depth + x + z, data = diamond_sample_log)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57253 -0.08124 -0.00114  0.08368  0.43240
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   6.34656    0.77589   8.180 8.81e-16 ***
## log_carat     1.76092    0.06519  27.011  < 2e-16 ***
## cutGood       0.08926    0.02714   3.289  0.00104 **
## cutVery Good  0.13282    0.02621   5.067 4.83e-07 ***
## cutPremium    0.12616    0.02654   4.755 2.29e-06 ***
## cutIdeal      0.15711    0.02599   6.045 2.12e-09 ***
## claritySI2    0.45195    0.04086  11.060  < 2e-16 ***
## claritySI1    0.61110    0.04081  14.974  < 2e-16 ***
## clarityVS2    0.75230    0.04083  18.426  < 2e-16 ***
## clarityVS1    0.83048    0.04149  20.018  < 2e-16 ***
## clarityVVS2   0.99966    0.04222  23.677  < 2e-16 ***
## clarityVVS1   1.01646    0.04379  23.212  < 2e-16 ***
## clarityIF     1.09085    0.04901  22.259  < 2e-16 ***
## colorE       -0.04718    0.01576  -2.993  0.00283 **
## colorF       -0.09589    0.01606  -5.971 3.30e-09 ***
## colorG       -0.13106    0.01561  -8.396  < 2e-16 ***
## colorH       -0.23909    0.01642 -14.563  < 2e-16 ***
## colorI       -0.38460    0.01827 -21.054  < 2e-16 ***
## colorJ       -0.49079    0.02092 -23.459  < 2e-16 ***
## depth         0.01768    0.01210   1.462  0.14411
## x             0.29582    0.11986   2.468  0.01376 *
## z            -0.38066    0.19426  -1.960  0.05034 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1301 on 976 degrees of freedom
## Multiple R-squared:  0.9844, Adjusted R-squared:  0.984
## F-statistic:  2929 on 21 and 976 DF,  p-value: < 2.2e-16
```

```
# AIC comparison across candidate models
aic_table <- AIC(model_log, model_log_R2, model_full, model_aic)
print(aic_table)
```

```
##              df       AIC
## model_log     3   138.853
## model_log_R2 20 -1206.681
## model_full   25 -1212.456
## model_aic    23 -1215.192
```

The AIC-selected model has a minimized AIC of -1,215.19 and uses the formula:

```
log_price ~ log_carat + cut + clarity + color + depth + x + z
```

2) Multicollinearity — Variance Inflation Factor (VIF)

```
# Test VIF of AIC model
vif(model_aic)
```

```
##                  GVIF Df GVIF^(1/(2*Df))
```

```
## log_carat   90.297684  1        9.502509
## cut          1.525012  4        1.054166
## clarity      1.479220  7        1.028360
## color        1.379716  6        1.027186
## depth       17.805722  1        4.219683
## x         1135.864995  1       33.702596
## z         1153.750447  1       33.966902
```

```
# VIF for x and z of 33.70 and 33.96 indicate severe collinearity
# VIF for log_carat of 9.50 indicates heavy collinearity, likely from the inclusion of x and z
# Should consider removing x and z to limit collinearity

# Test VIF of full model
vif(model_full)
```

```
##                  GVIF Df GVIF^(1/(2*Df))
## log_carat   97.369850  1        9.867616
## cut          2.504226  4        1.121590
## clarity      1.538178  7        1.031235
## color        1.420946  6        1.029710
## depth       35.433845  1        5.952633
## table        1.771540  1        1.330992
## x         1141.154509  1       33.780979
## y         1237.400302  1       35.176701
## z         2137.527828  1       46.233406
```

```
# Severe collinearity for x, y, z, and log_carat
# Moderate collinearity for depth

# Test VIF of maximized R^2 model
vif(model_log_R2)
```

```
##               GVIF Df GVIF^(1/(2*Df))
## log_carat 1.383791  1        1.176346
## cut       1.163132  4        1.019069
## clarity   1.392899  7        1.023953
## color     1.290697  6        1.021493
```

```
# All VIF values are 1.17 or less, indicating no collinearity
# This model should be chosen
```

VIF is shown as GVIF^(1/(2*Df)).

Collinearity thresholds: 1-2 = acceptable | 2–5 = moderate | > 5 = severe

- **AIC model:** x and z have VIF values of 33.70 and 33.96, indicating severe collinearity. VIF for log_carat of 9.50 also indicates indicates collinearity, likely from the inclusion of x and z. Should consider removing x and z to limit this.
- **Full model:** Severe collinearity for x, y, z, log_carat, and depth.
- **Maximized R^2 model**: All VIF values are 1.17 or less, indicating acceptable collinearity. This confirms the final model choice.

3) Confidence and Prediction Intervals

Combination 1: 1.0 carat, Ideal cut, VVS1 clarity, D color

```
# For a diamond with Carat = 1, an Ideal cut, VVS1 clarity, and D color
diamond_1 <- data.frame(log_carat = log(1.0),
```

```
                          cut = factor("Ideal",
                                       levels = c("Fair","Good",
                                                  "Very Good","Premium",
                                                  "Ideal")),
                          clarity = factor("VVS1",
                                           levels = c("I1","SI2","SI1",
                                                      "VS2","VS1",
                                                      "VVS2","VVS1","IF")),
                          color = factor("D",
                                         levels = c("D","E","F","G","H",
                                                    "I","J")))

# Confidence Interval (back-transformed to USD)
exp(predict(model_log_R2, newdata = diamond_1, interval = "confidence",
        level = 0.95))
```

```
##        fit      lwr      upr
## 1 8177.732 7817.498 8554.566
```

```
# Prediction Interval (back-transformed to USD)
exp(predict(model_log_R2, newdata = diamond_1, interval = "prediction",
        level = 0.95))
```

```
##        fit      lwr      upr
## 1 8177.732 6301.377 10612.81
```

- **Confidence Interval:** 95% confident the true mean price for all 1 carot, Ideal, VVS1, D diamonds falls between $7,817.50 and $8,554.57.
- **Prediction Interval:** 95% confident the price for one specific such diamond falls between $6,301.38 and $10,612.81.

Combination 2: 2.0 carat, Fair cut, IF clarity, J color

```
diamond_2 <- data.frame(log_carat = log(2.0),
                          cut = factor("Fair",
                                       levels = c("Fair","Good",
                                                  "Very Good","Premium",
                                                  "Ideal")),
                          clarity = factor("IF",
                                           levels = c("I1","SI2","SI1",
                                                      "VS2","VS1",
                                                      "VVS2","VVS1","IF")),
                          color = factor("J",
                                         levels = c("D","E","F","G","H",
                                                    "I","J")))

# Confidence Interval
exp(predict(model_log_R2, newdata = diamond_2, interval = "confidence",
        level = 0.95))
```

```
##        fit      lwr      upr
## 1 16636.99 15372.91 18005.01
```

```
# Prediction Interval
exp(predict(model_log_R2, newdata = diamond_2, interval = "prediction",
        level = 0.95))
```

```
##        fit   lwr     upr
## 1 16636.99 12718 21763.6
```

- **Confidence Interval:** 95% confident the true mean price for all 2 carat, Fair, IF, J diamonds falls between $15,372.91 and $18,005.01.
- **Prediction Interval:** 95% confident the price for one specific such diamond falls between $12,718.00 and $21,764.60.

---

4) Report Summary

The aim of this project was to model diamond prices using the provided Diamonds dataset, identify the main contributing variables for price, and create the best linear regression model.

Part 1

A random sample of 1,000 diamonds was used from 53,943 observations. One observation was later identified as an outlier caused by data entry error, and was therefore removed.

For the quantitative variables, carat, price, x, y, and z, were all right skewed, indicating the sample was filled with small and relatively cheaper diamonds. Depth and table percentage were approximately normally distributed. For the categorical variables, the most common observations were an ideal cut, mid-range color grades, and slightly to very slightly included clarity grades.

When briefly looking into correlation, it was clear that carat was strongly correlated with depth (x). Later, it would become apparent that carat was correlated with all other quantitative variables, depth %, table %, x, y, and z.

Part 2

The simple linear regression model (price ~ carat) produced an adjusted R^2 of 0.8556, but violated normality of residuals and homoscedasticity. A log transformation on both variables corrected these violations and improved the adjusted R^2 to 0.9368.

After adding additional variables to maximize adjusted R^2, the value improved to 0.9838593. The variables added were cut, clarity, and color. This model would eventually be chosen as the best and final model.

Part 3

The minimized AIC model produced an AIC of -1215.192, and used all variables except for table and width (y). This confirmed that log_cut, cut, clarity, and color belong in the final model.

VIF values in the minimized AIC model indicated severe collinearity between x, z, and log_carat. The full model showed even more severe signs of collinearity, also between x, y, z, and log_carat. Both models showed moderate collinearity for depth percentage.

For the maximized adjusted R^2 model, VIF values indicated no collinearity between variables. Therefore, this model was chosen as the best and final.

Final Model

```
summary(model_log_R2)
```

```
##
## Call:
## lm(formula = log_price ~ log_carat + cut + clarity + color, data = diamond_sample_log)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.55413 -0.08349  0.00041  0.08052  0.41409
##
```

```
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.817025   0.045114 173.272  < 2e-16 ***
## log_carat     1.877198   0.008117 231.276  < 2e-16 ***
## cutGood       0.098560   0.026515   3.717 0.000213 ***
## cutVery Good  0.146318   0.024695   5.925 4.32e-09 ***
## cutPremium    0.151450   0.024652   6.144 1.17e-09 ***
## cutIdeal      0.174881   0.024398   7.168 1.50e-12 ***
## claritySI2    0.452808   0.040858  11.082  < 2e-16 ***
## claritySI1    0.607237   0.040646  14.940  < 2e-16 ***
## clarityVS2    0.751236   0.040829  18.400  < 2e-16 ***
## clarityVS1    0.829189   0.041505  19.978  < 2e-16 ***
## clarityVVS2   1.001561   0.042312  23.671  < 2e-16 ***
## clarityVVS1   1.017264   0.043893  23.176  < 2e-16 ***
## clarityIF     1.089941   0.049097  22.200  < 2e-16 ***
## colorE       -0.050764   0.015792  -3.214 0.001349 **
## colorF       -0.099013   0.016120  -6.142 1.18e-09 ***
## colorG       -0.132684   0.015619  -8.495  < 2e-16 ***
## colorH       -0.239505   0.016410 -14.595  < 2e-16 ***
## colorI       -0.382922   0.018161 -21.085  < 2e-16 ***
## colorJ       -0.488757   0.020903 -23.382  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1308 on 979 degrees of freedom
## Multiple R-squared:  0.9842, Adjusted R-squared:  0.9839
## F-statistic:  3377 on 18 and 979 DF,  p-value: < 2.2e-16
```

$$\log(\text{price}) = \beta_0 + \beta_1 \cdot \log(\text{carat}) + \beta_2 \cdot \text{cut} + \beta_3 \cdot \text{clarity} + \beta_4 \cdot \text{color}$$

Adjusted $R^2 = 0.9839$

Key Takeaways

The physical dimensions x, y, and z are redundant when carat is included in the model. Intuitively weight increases with size, so the model understandably only needs to account for the diamond's weight. Table % and depth % also provide minimal explanatory power when carat is included, as shown by the adjusted R^2 decreasing when each variable is added.

The four variables of carat, cut, clarity, and color explain roughly 98% of price variability. This aligns with industry consensus, which prices diamonds based on the four C's.